

Perpendicular lines and planes in space

Two lines is enough — the test that turns an infinite condition into a finite check.

LESSON 33–34

2 × 40 MIN

GEOMETRY 10, §14

EXTENSION

LESSON CLOCK

10'

18'

18'

16'

13'

5'

The definition	10 min
The test	18 min
Two lemmas	18 min
Construction and uniqueness	16 min
Practice	13 min
Homework	5 min

LEARNING OBJECTIVES

- Define a line perpendicular to a plane.
- State and apply the test for perpendicularity of a line and a plane.
- Prove the two lemmas on parallel lines and perpendicular planes.
- Construct the perpendicular from a point to a plane.

TEXTBOOK

National	pp. 147–158
Cambridge	Core & Extended, pp. 323–328
Workbook	Exercise 14.1

01

Explanation

The definition

DEFINITION

A line is **perpendicular to a plane** when it is perpendicular to **every** line lying in that plane.

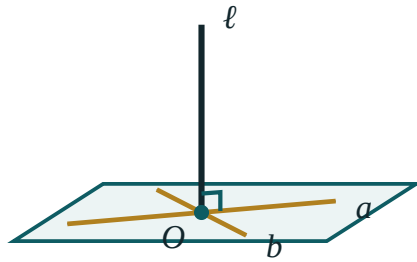
As a definition that is unusable — there are infinitely many lines to check. The value of the next theorem is that it reduces “every” to “two”.

The line is also called a **normal** to the plane, and the point where it meets the plane is the **foot** of the perpendicular.

The test

TEST FOR A LINE PERPENDICULAR TO A PLANE

If a line is perpendicular to **two intersecting** lines of a plane, it is perpendicular to the whole plane.



$l \perp a$ and $l \perp b$, with a and b intersecting at O
therefore l is perpendicular to the whole plane

Two intersecting lines are enough. A third would be redundant; one would not be enough.

INTERSECTING, NOT PARALLEL

A line perpendicular to two **parallel** lines of a plane need not be perpendicular to the plane: a pencil laid across two rails of a railway is perpendicular to both, and lies flat in the plane. The two directions must be genuinely different.

In a cube, $AA_1 \perp AB$ and $AA_1 \perp AD$, and AB and AD intersect at A . So $AA_1 \perp ABCD$, and therefore $AA_1 \perp AC$ and $AA_1 \perp BD$ as well — results that would otherwise each need their own proof.

Two lemmas

Lemma 1. If one of two parallel lines is perpendicular to a plane, so is the other.

Lemma 2. If two lines are perpendicular to the same plane, they are parallel.

Together these say: **all the normals of a plane are parallel to one another**, and any line parallel to a normal is itself a normal. The direction perpendicular to a plane is unique.

$$a \perp \alpha \text{ and } b \perp \alpha \Rightarrow a \parallel b$$

COMPARE WITH THE PLANE CASE

In the plane, two lines perpendicular to the same line are parallel — the same statement, one dimension down. Lemma 2 is its space version, and it is proved the same way.

Construction and uniqueness

Theorem. Through any point there passes exactly one line perpendicular to a given plane, and exactly one plane perpendicular to a given line.

The distance from a point to a plane is the length of that perpendicular segment — and by the argument of the next lesson, it is the shortest of all segments from the point to the plane.

IN A CUBE $ABCD A_1 B_1 C_1 D_1$	PERPENDICULAR TO
AA_1	$ABCD$ and $A_1 B_1 C_1 D_1$
AB	$ADD_1 A_1$ and $BCC_1 B_1$
AD	$ABB_1 A_1$ and $DCC_1 D_1$

02 Worked examples

EXAMPLE 1 In a cube, prove $AA_1 \perp AC$.

- ① $AA_1 \perp AB$ and $AA_1 \perp AD$.
Adjacent edges of square faces.
- ② AB and AD intersect at A and lie in $ABCD$.
- ③ By the test, $AA_1 \perp ABCD$.
- ④ $AC \subset ABCD$.
By the definition.

ANSWER

$AA_1 \perp AC$

EXAMPLE 2 A line is perpendicular to two parallel lines of a plane. Must it be perpendicular to the plane?

- 1 The test demands two **intersecting** lines.
- 2 A line inside the plane, crossing both, is perpendicular to both.
- 3 It is not perpendicular to the plane.

ANSWER

No

EXAMPLE 3 Two lines a and b are perpendicular to the plane α . What can be said?

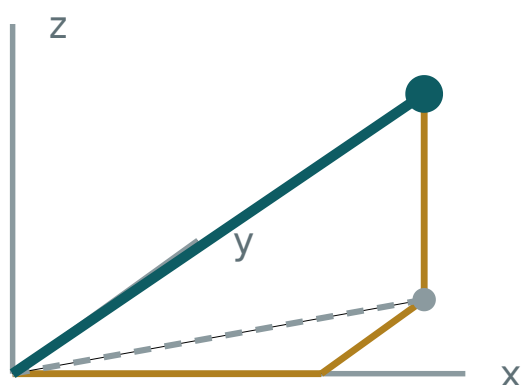
- 1 Lemma 2 applies.
- 2 $a \parallel b$
- 3 They determine a plane, perpendicular to α .

ANSWER

 $a \parallel b$ **03 Interactive model**

Stand a pencil on the desk and check it against two rulers laid across it.

A normal to a plane



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Perpendicular to a plane	Tekislikka perpendikulyar	Перпендикуляр к плоскости
Test for perpendicularity	Perpendikulyarlik alomati	Признак перпендикулярности
Two intersecting lines	Kesishuvchi ikki chiziq	Две пересекающиеся прямые
Foot of the perpendicular	Perpendikulyar asosi	Основание перпендикуляра
Normal to a plane	Tekislik normali	Нормаль к плоскости
Unique perpendicular	Yagona perpendikulyar	Единственный перпендикуляр
Lemma	Lemma	Лемма
Construction	Yasash	Построение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A line $\perp$ to a plane is $\perp$ to:

one line of it

two lines of it

every line of it

no line of it

The test needs:

one line

two intersecting lines

two parallel lines

three lines

Two lines $\perp$ to the same plane are:

perpendicular

parallel

skew

equal

If $a \parallel b$ and $a \perp \alpha$ then:

$$b \parallel \alpha$$

$$b \perp \alpha$$

$$b \subset \alpha$$

nothing

In a cube AA_1 is perpendicular to:

$$ABB_1A_1$$

$$ABCD$$

$$ADD_1A_1$$

all of them

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Define a line perpendicular to a plane

ANSWER $\perp$ to every line of the plane

2. State the test

ANSWER $\perp$ to two intersecting lines of the plane

3. Is $AA_1 \perp ABCD$ in a cube?

ANSWER yes

4. Is $AA_1 \perp AC$?

ANSWER yes

5. Two lines $\perp$ to the same plane are

ANSWER parallel

6. Is one line enough for the test?

ANSWER no

7. Are two parallel lines enough?

ANSWER no

Medium · the standard question

7 PROBLEMS

1. Prove $AA_1 \perp BD$ in a cube

ANSWER $AA_1 \perp ABCD$ and $BD \subset ABCD$

2. Which faces is AB perpendicular to?

ANSWER ADD_1A_1 and BCC_1B_1

3. If $a \perp \alpha$ and $b \perp \alpha$, what is the angle between a and b ?

ANSWER 0° — parallel

4. How many perpendiculars to a plane through a point?

ANSWER 1

5. How many planes $\perp$ to a line through a point?

ANSWER 1

6. Give a counterexample to "two parallel lines suffice"

ANSWER A line inside the plane crossing two rails

7. In a cube, is $AC \perp BD$?

ANSWER yes — diagonals of a square

Hard · stretch and reason

7 PROBLEMS

1. Prove Lemma 1

ANSWER Translate the perpendicularity along the parallel

2. Prove Lemma 2

ANSWER Otherwise two planes through one point would be perpendicular to one line

3. In a cube, prove $BD \perp AA_1C_1C$ ANSWER $BD \perp AC$ and $BD \perp AA_1$ 4. In a cube, prove $AC_1 \perp BD$ ANSWER $BD \perp AA_1C_1C$ contains AC_1 5. A regular pyramid: prove the axis is $\perp$ to the baseANSWER It is $\perp$ to two base diagonals6. If $a \perp \alpha$ and $\alpha \parallel \beta$, is $a \perp \beta$?

ANSWER yes

7. Can a line be perpendicular to two intersecting planes?

ANSWER no — the normals would be parallel

07 Homework**Homework — 5 tasks**

Every proof must name the two intersecting lines used in the test.

1. In cube $ABCD A_1 B_1 C_1 D_1$, prove $DD_1 \perp AC$.
2. List every face of a cube that BC is perpendicular to.
3. Prove that if two lines are perpendicular to the same plane they are parallel.
4. Give a counterexample showing that two parallel lines are not enough for the test.
5. In a cube, prove $BD \perp AA_1C_1C$ and deduce $BD \perp AC_1$.

The perpendicular, the oblique line and distance in space

Three lessons on the shortest segment — and the three theorems that compare obliques with it.

LESSON 35–37

3 × 40 MIN

GEOMETRY 10, §15

EXTENSION

LESSON CLOCK

14'

22'

22'

26'

26'

10

Three objects, one picture	14 min
The shortest segment	22 min
Comparing obliques	22 min
Distances	26 min
Practice	26 min
Homework	10 min

LEARNING OBJECTIVES

- Distinguish the perpendicular, an oblique line and its projection.
- Prove that the perpendicular is the shortest segment to a plane.
- Compare obliques by their projections.
- Compute distances from a point to a plane and between parallel planes.

TEXTBOOK

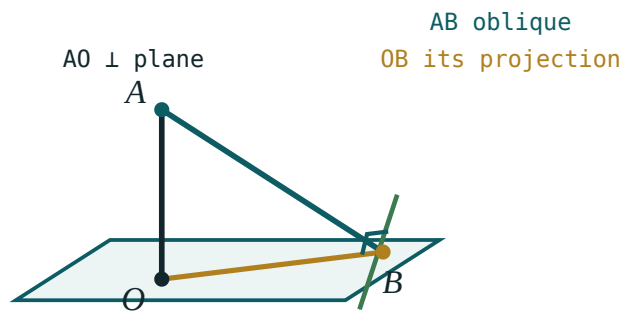
National	pp. 159–176
Cambridge	Core & Extended, pp. 329–334
Workbook	Exercise 15.1

01

Explanation

Three objects, one picture

From a point A not in a plane α , drop the **perpendicular** AO . Any other segment AB from A to the plane is an **oblique**, and OB is its **projection**.



a line of the plane through B is $\perp$ AB
exactly when it is $\perp$ OB

*The perpendicular, the oblique and the projection form
a right triangle in a vertical plane.*

THE RIGHT TRIANGLE

$\triangle AOB$ always has its right angle at O , because $AO \perp \alpha$ and $OB \subset \alpha$. So $AB^2 = AO^2 + OB^2$ — Pythagoras — and every theorem below is a consequence.

The shortest segment

Theorem 1. The perpendicular is shorter than any oblique from the same point.

Proof. $AB^2 = AO^2 + OB^2 > AO^2$ whenever $OB > 0$. ■

DEFINITION OF DISTANCE

The **distance from a point to a plane** is the length of the perpendicular. Theorem 1 is what makes that definition sensible: it is genuinely the shortest.

The same argument, one dimension down, is why the distance from a point to a line in the plane is measured along the perpendicular. Nothing new is being asserted — only extended.

Comparing obliques

Theorem 2. Equal obliques have equal projections, and equal projections give equal obliques.

Theorem 3. Of two obliques, the one with the longer projection is longer.

$$OB_1 > OB_2 \Leftrightarrow AB_1 > AB_2$$

Both follow immediately from $AB^2 = AO^2 + OB^2$: with AO fixed, AB grows exactly as OB does.

A USEFUL COROLLARY

All the obliques of a given length from A land on a **circle** centred at O — because they all have the same projection length. That circle is how a cone meets a plane, and it is the reason a regular pyramid's lateral edges are equal.

The **angle between the oblique and the plane** is $\angle ABO$, the angle with its own projection — and it is the smallest angle AB makes with any line of the plane.

Distances

DISTANCE	HOW IT IS MEASURED
point to plane	the perpendicular from the point
line parallel to a plane, to that plane	from any point of the line — it is constant
between parallel planes	from any point of one to the other
between skew lines	the common perpendicular (Quarter II)

Example. In a cube of edge a , the distance from A to the plane $A_1B_1C_1D_1$ is a ; from A to the plane BDD_1B_1 is half the diagonal AC , that is $\frac{a\sqrt{2}}{2}$.

02 Worked examples

EXAMPLE 1 From a point 10 cm above a plane, two obliques have projections 6 and 8 cm. Find their lengths.

$$1 \quad AB_1 = \sqrt{100 + 36} = \sqrt{136} \approx 11.66$$

$$2 \quad AB_2 = \sqrt{100 + 64} = \sqrt{164} \approx 12.81$$

3 The longer projection gives the longer oblique ✓

ANSWER

≈ 11.7 and ≈ 12.8 cm

EXAMPLE 2 An oblique of length 13 makes 30° with the plane. Find the perpendicular and the projection.

1 The angle is $\angle ABO$.

$$2 \quad AO = 13 \sin 30^\circ = 6.5$$

$$3 \quad OB = 13 \cos 30^\circ \approx 11.26$$

ANSWER

$AO = 6.5$, $OB \approx 11.3$

EXAMPLE 3 Cube of edge 6. Find the distance from A to the plane BDD_1B_1 .

- ① $AC \perp BD$ and $AC \perp BB_1$.
So $AC \perp BDD_1B_1$.

- ② The foot is the centre of the base.

- ③ $AC = 6\sqrt{2}$, half is $3\sqrt{2}$.

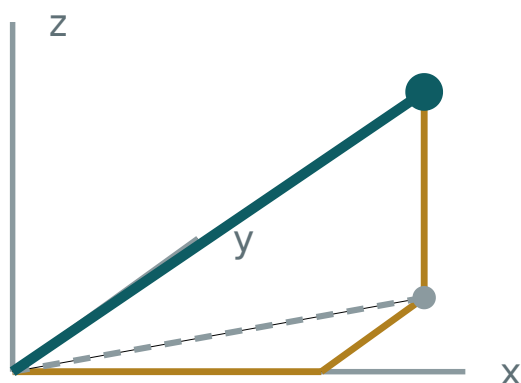
ANSWER

$$3\sqrt{2} \approx 4.24$$

03 Interactive model

Hold a stick to the table and rotate it — the shadow lengthens exactly as the stick does.

Perpendicular and oblique



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Perpendicular	Perpendikulyar	Перпендикуляр
Oblique line	Og'ma	Наклонная
Projection of an oblique	Og'ma proyeksiyasi	Проекция наклонной
Foot of the perpendicular	Perpendikulyar asosi	Основание перпендикуляра
Distance from a point to a plane	Nuqtadan tekislikkacha masofa	Расстояние от точки до плоскости
Distance between parallel planes	Parallel tekisliklar orasidagi masofa	Расстояние между плоскостями
Shortest segment	Eng qisqa kesma	Кратчайший отрезок
Equal obliques	Teng og'malar	Равные наклонные
Angle with the plane	Tekislik bilan burchak	Угол с плоскостью

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The distance from a point to a plane is measured along:

any segment

the perpendicular

the oblique

the projection

AB^2 equals:

$AO^2 - OB^2$

$AO^2 + OB^2$

$AO \cdot OB$

$AO + OB$

Equal obliques have:

equal projections

unequal projections

no projection

equal angles only

The longer projection belongs to:

the shorter oblique

the longer oblique

the perpendicular

neither

The angle between an oblique and a plane is measured to:

any line of the plane

its projection

the normal

the edge

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Perpendicular 3, projection 4: the oblique

ANSWER 5

2. Perpendicular 5, projection 12: the oblique

ANSWER 13

3. Oblique 10, perpendicular 6: the projection

ANSWER 8

4. Oblique 25, projection 24: the perpendicular

ANSWER 7

5. Distance from a point to a plane is measured along

ANSWER the perpendicular

6. Equal projections give

ANSWER equal obliques

7. Distance from A to $A_1B_1C_1D_1$ in a cube of edge 6

ANSWER 6

Medium · the standard question

7 PROBLEMS

1. Point 10 above; projections 6 and 8: the obliques

ANSWER $\sqrt{136}$ and $\sqrt{164}$

2. Oblique 13 at 30° : perpendicular and projection

ANSWER 6.5, ≈ 11.3

3. Oblique 20 at 60° : perpendicular

ANSWER $10\sqrt{3} \approx 17.3$

4. Cube edge 6: distance from A to BDD_1B_1

ANSWER $3\sqrt{2}$

5. Perpendicular 8, oblique 17: the projection

ANSWER 15

6. Two obliques 15 and 17 with perpendicular 9: the projections

ANSWER 12 and $\sqrt{208}$

7. Angle of an oblique with perpendicular 6 and projection 8

ANSWER $\approx 36.9^\circ$

Hard · stretch and reason

7 PROBLEMS

1. Cube edge a : distance from A to the plane A_1BD

ANSWER $\frac{a\sqrt{3}}{3}$

2. Cube edge 6: distance from B to the plane ACC_1A_1

ANSWER $3\sqrt{2}$

3. A point is 12 from a plane; an oblique makes 40° . Find it

ANSWER ≈ 18.7

4. Three equal obliques from A : what is the foot of the perpendicular?

ANSWER The circumcentre of the three feet

5. A regular pyramid: prove the lateral edges are equal

ANSWER Equal projections — the base circumradius

6. Distance between the parallel planes $ABCD$ and $A_1B_1C_1D_1$ in a cube of edge 9

ANSWER 9

7. Prove that the angle with the projection is the smallest angle with any line of the plane

ANSWER Compare cosines in the right triangles

07 Homework

Homework — 6 tasks

Draw the right triangle AOB separately for every question.

- From a point 8 cm above a plane, two obliques have projections 6 and 15 cm. Find both lengths.
- An oblique of 26 cm makes 30° with a plane. Find the perpendicular and the projection.
- Cube of edge 8: find the distance from A to the plane BDD_1B_1 .
- A perpendicular is 9 and an oblique is 41. Find the projection and the angle with the plane.
- Prove that the perpendicular is shorter than every oblique from the same point.

6. Explain why the lateral edges of a regular pyramid are equal.

The theorem of the three perpendiculars

One theorem that turns every “is this angle right?” question in space into a question about a shadow.

LESSON 38–39

2 × 40 MIN

GEOMETRY 10, §16

EXTENSION

LESSON CLOCK

11'

17'

17'

17'

12'

6'

The statement	11 min
The proof	17 min
Applications	17 min
Distance to a line	17 min
Practice	12 min
Homework	6 min

LEARNING OBJECTIVES

- State the theorem of the three perpendiculars and its converse.
- Apply it to prove perpendicularity in a solid.
- Use it to find the distance from a point to a line in space.
- Recognise the three perpendiculars in a given figure.

TEXTBOOK

National	pp. 177–188
Cambridge	Core & Extended, pp. 335–338
Workbook	Exercise 16.1

01

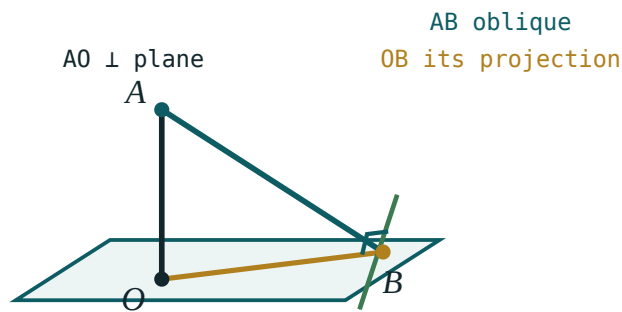
Explanation

The statement

THEOREM OF THE THREE PERPENDICULARS

Let $AO \perp \alpha$ with foot O , let AB be an oblique with projection OB , and let c be a line in α passing through B . Then

$c \perp AB \Leftrightarrow c \perp OB$



a line of the plane through B is $\perp$ AB
exactly when it is $\perp$ OB

*AO is the perpendicular, AB the oblique, OB its
projection. A line of the plane is $\perp$ to one exactly when
it is $\perp$ to the other.*

The three perpendiculars of the name are $AO \perp \alpha$, $c \perp OB$ and $c \perp AB$ — any two give the third.

The proof

Forward direction. Suppose $c \perp OB$. Since $AO \perp \alpha$ and $c \subset \alpha$, also $c \perp AO$. So c is perpendicular to two intersecting lines OB and AO of the plane AOB . By the test of Lesson 33–34, $c \perp AOB$, and therefore $c \perp AB$, since $AB \subset AOB$. ■

Converse. The same argument with AB and AO in place of OB and AO . ■

THE PATTERN TO REMEMBER

Prove perpendicularity to a **plane**, then read off perpendicularity to any line in it. Almost every space perpendicularity proof in this course has that shape.

Applications

In a cube. Is $AC_1 \perp BD$? Project AC_1 onto the base: its projection is AC . Since $BD \perp AC$ (diagonals of a square) and $BD \subset ABCD$, the theorem gives $BD \perp AC_1$ at once — no coordinates, no vectors, two lines of work.

In a pyramid. A regular square pyramid $SABCD$ with centre O . Is $SM \perp AB$, where M is the midpoint of AB ? The projection of SM is OM , which is perpendicular to AB by symmetry. So yes — and SM is the slant height.

THE LINE MUST LIE IN THE PLANE AND PASS THROUGH THE FOOT

c must be in α and meet OB at B . A line of the plane not through B is not covered by the theorem — translate it first.

The distance from a point to a line in space

To find the distance from A to a line c lying in a plane α , when $AO \perp \alpha$:

1. Drop $OB \perp c$ inside the plane — an ordinary plane construction.
2. By the theorem, $AB \perp c$ too.
3. So AB is the required distance, and $AB = \sqrt{AO^2 + OB^2}$.

Example. A is 12 above the plane; the foot O is 5 from the line c . Then the distance from A to c is $\sqrt{144 + 25} = 13$.

02 Worked examples

EXAMPLE 1 In a cube, prove $AC_1 \perp BD$.

- 1 $CC_1 \perp ABCD$, so AC is the projection of AC_1 .
- 2 $BD \perp AC$
Diagonals of a square.
- 3 $BD \subset ABCD$ and passes through the foot.
- 4 By the theorem of the three perpendiculars.

ANSWER

$$AC_1 \perp BD$$

EXAMPLE 2 A is 12 cm above a plane, and its foot O is 5 cm from a line c in the plane. Find the distance from A to c .

① Drop $OB \perp c$ in the plane; $OB = 5$.

② By the theorem, $AB \perp c$.

③ $AB = \sqrt{144 + 25} = 13$

ANSWER

13 cm

EXAMPLE 3 A regular square pyramid has base edge 10 and height 12. Find the slant height.

① $OM = 5$, the apothem of the base.

② $SO \perp ABCD$, so OM is the projection of SM .

③ $OM \perp AB$, so $SM \perp AB$.

It is the slant height.

④ $SM = \sqrt{144 + 25} = 13$

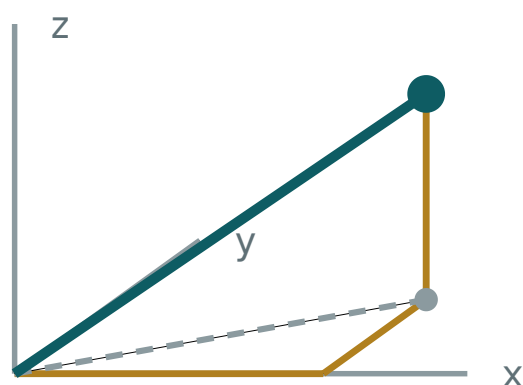
ANSWER

13

03 Interactive model

Point at the three perpendiculars on a real box before any calculation.

The three perpendiculars



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Theorem of the three perpendiculars	Uch perpendikulyar haqidagi teorema	Теорема о трёх перпендикулярах
Oblique line	Og'ma	Наклонная
Projection	Proyeksiya	Проекция
Converse theorem	Teskari teorema	Обратная теорема
Line in the plane	Tekislikdagi chiziq	Прямая в плоскости
Foot of the perpendicular	Perpendikulyar asosi	Основание перпендикуляра
Distance to a line	Chiziqqacha masofa	Расстояние до прямой
Necessary and sufficient	Zarur va yetarli	Необходимо и достаточно

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The theorem relates:

two obliques

an oblique and its projection

two planes

two normals

The line c must:

be anywhere

lie in the plane through the foot of the oblique

be perpendicular to the plane

be parallel to AO

In a cube $AC_1 \perp BD$ because:

$AC \perp BD$

$AA_1 \perp BD$

both

neither

If $AO = 12$ and $OB = 5$ then AB :

7

13

17

60

The theorem lets you replace a 3-D question with:

a 2-D question in the plane

a calculation

a guess

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. State the theorem of the three perpendiculars

ANSWER $c \perp AB \Leftrightarrow c \perp OB$

2. $AO = 3$, $OB = 4$; find AB

ANSWER 5

3. $AO = 12$, $OB = 5$; find AB

ANSWER 13

4. $AB = 25$, $OB = 24$; find AO

ANSWER 7

5. Is $AC_1 \perp BD$ in a cube?

ANSWER yes

6. The projection of AC_1 on the base

ANSWER AC

7. Where must c lie?

ANSWER in the plane, through B

Medium · the standard question

7 PROBLEMS

1. Prove $AC_1 \perp BD$ in a cube

ANSWER Three perpendiculars, with projection AC

2. A is 12 above; O is 5 from c . Distance to c

ANSWER 13

3. Regular pyramid, base 10, height 12: slant height

ANSWER 13

4. Regular pyramid, base 16, height 6: slant height

ANSWER 10

5. A is 8 above; distance to a line 15 from O

ANSWER 17

6. In a cube, is $BD_1 \perp AC$?

ANSWER yes

7. Prove $SM \perp AB$ in a regular pyramid

ANSWER $OM \perp AB$ and the theorem

Hard · stretch and reason

7 PROBLEMS

1. Cube edge a : prove $AC_1 \perp BDA_1$

ANSWER Three perpendiculars twice

2. Regular pyramid, base 12, lateral edge 10: the slant height

ANSWER 8

3. Same pyramid: the height

ANSWER $\sqrt{82} \approx 9.06$

4. A point 15 above a plane; two lines in the plane at distances 8 and 20 from O . Distances from A

ANSWER 17 and 25

5. Prove the converse of the theorem

ANSWER Same argument with AB in place of OB

6. In a cube, find the distance from A to the line B_1C_1 for edge 6

ANSWER $6\sqrt{2}$

7. Why must c pass through B ?

ANSWER Otherwise it is not in the plane AOB

07 Homework

Homework — 5 tasks

Name the perpendicular, the oblique and the projection in every solution.

1. State the theorem of the three perpendiculars and prove the forward direction.
2. In a cube, prove $BD_1 \perp AC$.
3. A is 9 cm above a plane and its foot is 12 cm from a line c . Find the distance from A to c .
4. A regular square pyramid has base edge 14 and height 24. Find the slant height and the lateral edge.

5. Explain in three sentences why the theorem reduces a space question to a plane one.

Perpendicularity of planes

The dihedral angle, its linear angle, and the test that makes two planes perpendicular.

LESSON 40–42

3 × 40 MIN

GEOMETRY 10, §17

EXTENSION

LESSON CLOCK

14'

22'

22'

26'

26'

10

The dihedral angle	14 min
The linear angle	22 min
The test	22 min
Computing dihedral angles	26 min
Practice	26 min
Homework	10 min

LEARNING OBJECTIVES

- Define a dihedral angle and construct its linear angle.
- State and apply the test for perpendicularity of two planes.
- Compute a dihedral angle in a cube and in a pyramid.
- Use the properties of perpendicular planes in proofs.

TEXTBOOK

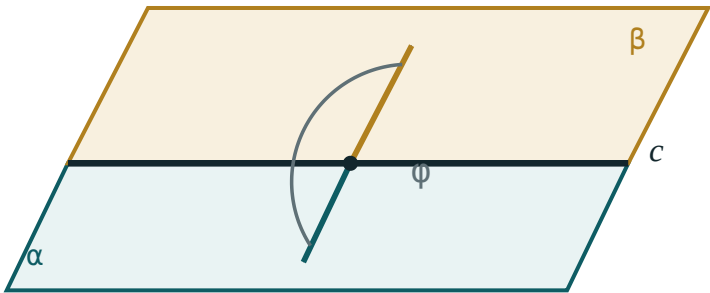
National	pp. 189–206
Cambridge	Core & Extended, pp. 339–344
Workbook	Exercise 17.1

01

Explanation

The dihedral angle

Two half-planes with a common edge divide space; the figure they form is a **dihedral angle**. Its **edge** is the common line; its **faces** are the two half-planes.



both arms are perpendicular to the edge c
the angle φ between them is the dihedral angle

*The edge, the two faces, and the linear angle
measured between arms perpendicular to the edge.*

THE LINEAR ANGLE

Take any point of the edge and draw, in each face, the ray perpendicular to the edge. The angle between those two rays is the **linear angle**, and it is what the size of the dihedral angle means.

BOTH ARMS PERPENDICULAR TO THE EDGE

Any other pair of rays gives a different, wrong angle. The linear angle does not depend on which point of the edge is chosen — but it does depend on both arms being perpendicular to the edge.

The test for perpendicular planes

DEFINITION AND TEST

Two planes are **perpendicular** when their dihedral angle is right. **Test:** if a plane contains a line perpendicular to another plane, the two planes are perpendicular.

$$\ell \subset \beta \text{ and } \ell \perp \alpha \Rightarrow \beta \perp \alpha$$

In a cube, $AA_1 \perp ABCD$ and $AA_1 \subset ABB_1A_1$, so $ABB_1A_1 \perp ABCD$. Every vertical face of a cube is perpendicular to the base, and one line proves it.

Two useful properties follow:

- If $\alpha \perp \beta$ and a line of α is perpendicular to their common edge, it is perpendicular to β .
- If two intersecting planes are each perpendicular to a third, their line of intersection is perpendicular to that third plane.

Computing dihedral angles

The whole difficulty is the construction of the linear angle. Three reliable routes:

1. **Drop a perpendicular** from a point of one face to the other plane, and join its foot to a point on the edge.
2. **Use the three perpendiculars theorem** to prove the arm you draw really is perpendicular to the edge.
3. **Choose the point on the edge conveniently** — usually a midpoint, so symmetry gives the arms for free.

Example. Cube of edge a : the dihedral angle along AB between the base $ABCD$ and the diagonal plane ABC_1D_1 . The arms are AD in the base and AD_1 in the other face,

both perpendicular to AB . The angle is $\angle DAD_1 = 45^\circ$, since ADD_1 is an isosceles right triangle.

In a pyramid

For a regular square pyramid with base edge $2b$ and height h , two dihedral angles are usually wanted:

ANGLE	ARMS	VALUE
lateral face to base	OM and SM	$\tan \theta = \frac{h}{b}$
lateral edge to base	OA and SA	$\tan \theta = \frac{h}{b\sqrt{2}}$

Base 10 , height 12 : the face angle has $\tan \theta = \frac{12}{5}$, so $\theta \approx 67.4^\circ$; the edge angle has

$\tan \theta = \frac{12}{5\sqrt{2}}$, so $\theta \approx 59.5^\circ$. The face is steeper than the edge — as the picture shows.

02 Worked examples

EXAMPLE 1 Cube of edge a . Find the dihedral angle along AB between $ABCD$ and ABC_1D_1 .

1 $AD \perp AB$ in the base.

2 $AD_1 \perp AB$ in the other face.
 $AB \perp ADD_1A_1$.

3 The linear angle is $\angle DAD_1$.

4 $AD = DD_1 = a$, so it is 45° .

ANSWER

45°

EXAMPLE 2 A regular square pyramid has base 10 and height 12. Find the dihedral angle between a lateral face and the base.

① $OM = 5$, the base apothem.

② $SO = 12$.

③ $\tan \theta = \frac{12}{5} = 2.4$

④ $\theta \approx 67.4^\circ$

ANSWER

$\approx 67.4^\circ$

EXAMPLE 3 Prove that $ABB_1A_1 \perp ABCD$ in a cube.

① $AA_1 \perp ABCD$

By the test of Lesson 33–34.

② $AA_1 \subset ABB_1A_1$

③ By the test for perpendicular planes.

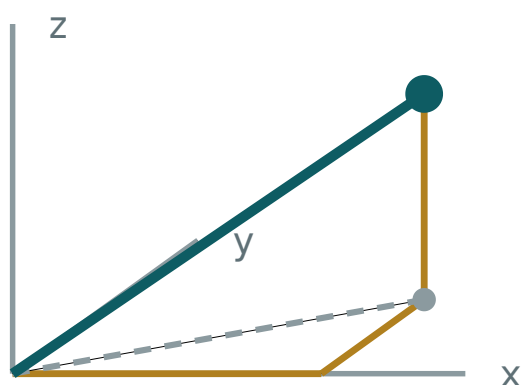
ANSWER

$ABB_1A_1 \perp ABCD$

03 Interactive model

Open a book on the desk: the spine is the edge, the pages the faces, and two pencils give the linear angle.

A dihedral angle



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Dihedral angle	Ikki yoqli burchak	Двугранный угол
Edge of a dihedral angle	Ikki yoqli burchak qirrasi	Ребро двугранного угла
Face (half-plane)	Yoq	Грань
Linear angle	Chiziqli burchak	Линейный угол
Perpendicular planes	Perpendikulyar tekisliklar	Перпендикулярные плоскости
Test for perpendicular planes	Tekisliklar perpendikulyarligi alomati	Признак перпендикулярности плоскостей
Right dihedral angle	To'g'ri ikki yoqli burchak	Прямой двугранный угол
Bisector plane	Bissektor tekislik	Биссекторная плоскость

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A dihedral angle is measured by:

any angle between the faces

its linear angle

the edge

the normal

The test for perpendicular planes: one contains:

a parallel line

a line perpendicular to the other

two points

the edge

In a cube, ABB_1A_1 and $ABCD$ are:

parallel

perpendicular

at 45°

skew

The dihedral angle along AB between $ABCD$ and ABC_1D_1 is:

Pyramid base 10, height 12: face-to-base angle:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. What is the edge of a dihedral angle?

ANSWER the common line of the two faces

2. How is a dihedral angle measured?

ANSWER by its linear angle

3. Both arms must be $\perp$ to

ANSWER the edge

4. State the test for perpendicular planes

ANSWER one contains a line $\perp$ to the other

5. Is $ABB_1A_1 \perp ABCD$ in a cube?

ANSWER yes

6. Is $ABCD \perp A_1B_1C_1D_1$?

ANSWER no — they are parallel

7. Dihedral angle along AB to ABC_1D_1

ANSWER 45°

Medium · the standard question

7 PROBLEMS

1. Pyramid base 10, height 12: face-to-base angle

ANSWER $\approx 67.4^\circ$

2. Same: edge-to-base angle

ANSWER $\approx 59.5^\circ$

3. Pyramid base 8, height 3: face-to-base angle

ANSWER $\approx 36.9^\circ$

4. Cube: dihedral angle along AB between $ABCD$ and ABB_1A_1

ANSWER 90°

5. Cube: dihedral angle between $ABCD$ and ACC_1A_1

ANSWER 90°

6. Prove $BCC_1B_1 \perp ABCD$

ANSWER $BB_1 \perp ABCD$ and lies in it

7. Regular tetrahedron: angle between a face and the base

ANSWER $\approx 70.5^\circ$

Hard · stretch and reason

7 PROBLEMS

1. Cube edge a : dihedral angle along AC between $ABCD$ and ACD_1

ANSWER $\approx 54.7^\circ$

2. Pyramid base 12, lateral edge 10: dihedral angle of a face with the base

ANSWER $\approx 43.3^\circ$

3. Prove: if two planes are each $\perp$ to a third, their line of intersection is $\perp$ to it

ANSWER It is $\perp$ to both edges

4. A regular tetrahedron of edge a : the dihedral angle between two faces

ANSWER $\arccos \frac{1}{3} \approx 70.5^\circ$

5. Cube: the angle between the planes ABC_1D_1 and A_1B_1CD

ANSWER 90°

6. A prism whose base is a rhombus with a 60° angle: the dihedral angles along the base edges

ANSWER 90° for a right prism

7. Prove that a plane $\perp$ to the edge of a dihedral angle cuts it in its linear angle

ANSWER Both cuts are $\perp$ to the edge

07 Homework

Homework — 6 tasks

Every dihedral answer must name the two arms and say why each is perpendicular to the edge.

1. Define a dihedral angle and explain how its linear angle is constructed.
2. Prove that $DCC_1D_1 \perp ABCD$ in a cube.
3. Cube of edge a : find the dihedral angle along AB between $ABCD$ and ABC_1D_1 .
4. A regular square pyramid has base edge 16 and height 6. Find the dihedral angle between a lateral face and the base.

5. The same pyramid: find the angle between a lateral edge and the base.
6. Prove that if two intersecting planes are each perpendicular to a third, their line of intersection is perpendicular to it.

Control work 3, and work on the mistakes

Perpendicularity in space in one paper — where naming the two intersecting lines is worth as much as the answer.

LESSON 43–44

2 × 40 MIN

GEOMETRY 10, NAZORAT ISHI 3

IGCSE REVIEW

LESSON CLOCK

3

35'

11'

20'

11'

Instructions	3 min
The paper	35 min
Answers	11 min
Rewrite	20 min
The test drill	11 min

LEARNING OBJECTIVES

- Apply the perpendicularity theorems under time.
- Quote a test by name and verify its conditions.
- Classify each lost mark and rewrite the solution.
- Consolidate the three perpendiculars theorem.

TEXTBOOK

National	pp. 207–210
Cambridge	Core & Extended, pp. 345–346
Workbook	Control paper G3

01

Explanation

The paper — 25 marks, 40 minutes

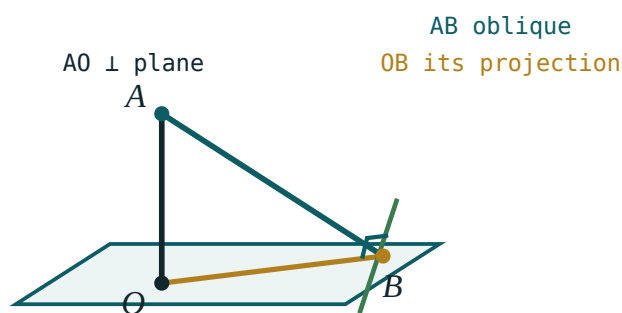
Q	TASK	MARKS	FROM
1	Prove $DD_1 \perp AC$ in a cube, quoting the test	4	L33–34
2	A point is 9 above a plane; two obliques have projections 12 and 40. Find both	4	L35–37
3	An oblique of 20 makes 30° with the plane. Find the perpendicular and the projection	4	L35–37
4	Prove $AC_1 \perp BD$ in a cube by the three perpendiculars	5	L38–39
5	A regular pyramid, base 10, height 12: find the slant height and the face-to-base dihedral angle	5	L38–42
6	Cube of edge 6: find the distance from A to the plane BDD_1B_1	3	L35–37

THREE MARKS FOR NAMING

Q1 gives one mark for naming the two intersecting lines, Q4 one for identifying the projection, Q5 one for saying why the arm is perpendicular to the edge. Three of twenty-five.

The three errors

ERROR	LOOKS LIKE	KIND
parallel lines used in the test	" $AA_1 \perp AB$ and $AA_1 \perp D_1C_1$, so $AA_1 \perp ABCD$ "	knowledge
wrong projection	the projection of AC_1 given as AB	method
arm not $\perp$ to the edge	a dihedral angle measured between any two rays	method
slant height and lateral edge confused	OM used where OA was needed	careless



a line of the plane through B is $\perp$ AB
exactly when it is $\perp$ OB

Questions 4 and 5 in one picture.

The test drill

Eight statements on the board. For each, the class names the **test or theorem** that settles it, in three seconds:

STATEMENT	TOOL
$AA_1 \perp ABCD$	test for line $\perp$ plane
$AC_1 \perp BD$	three perpendiculars
$ABB_1A_1 \perp ABCD$	test for perpendicular planes
$SM \perp AB$ in a pyramid	three perpendiculars
the shortest segment to a plane	the perpendicular

02 Worked examples

EXAMPLE 1 Model answer, Q4: prove $AC_1 \perp BD$ in a cube.

- 1 $CC_1 \perp ABCD$
So AC is the projection of AC_1 .
- 2 $BD \perp AC$
Diagonals of a square.
- 3 $BD \subset ABCD$ through the foot C .
- 4 Three perpendiculars.

ANSWER

$$AC_1 \perp BD$$

EXAMPLE 2 Model answer, Q5: pyramid base 10, height 12.

- 1 $OM = 5$
Half the base edge.
- 2 $SM = \sqrt{144 + 25} = 13$
The slant height.
- 3 $OM \perp AB$, and $SM \perp AB$ by the theorem.
The linear angle is $\angle SMO$.
- 4 $\tan \theta = \frac{12}{5} \Rightarrow \theta \approx 67.4^\circ$

ANSWER

$$SM = 13; \theta \approx 67.4^\circ$$

EXAMPLE 3 A learner used $AA_1 \perp AB$ and $AA_1 \perp D_1C_1$ to conclude $AA_1 \perp ABCD$. Name the mistake.

- ① D_1C_1 does not lie in $ABCD$.
- ② And it is parallel to AB , not intersecting.
Two failures at once.
- ③ Correct: use AB and AD .

ANSWER

The two lines must lie in the plane and intersect

03 Interactive model

Run the test drill before the rewrite, holding a real box.

Name the tool

To prove $AA_1 \perp ABCD$ use:

three perpendiculars

the test for line $\perp$ plane

the definition

Pythagoras

To prove $AC_1 \perp BD$ use:

three perpendiculars

the test for planes

the definition

measurement

Perpendicular 9, projection 12: the oblique is:

15

21

3

108

Oblique 20 at 30° : the perpendicular is:

Pyramid base 10, height 12: the slant height is:

Cube edge 6: distance from A to BDD_1B_1 :

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Test (criterion)	Alomat	Признак
Three perpendiculars	Uch perpendikulyar	Три перпендикуляра
Dihedral angle	Ikki yoqli burchak	Двугранный угол
Linear angle	Chiziqli burchak	Линейный угол
Justification	Asoslash	Обоснование
Careless error	E'tiborsizlik xatosi	Ошибка по невнимательности
Correction	Tuzatish	Исправление

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The test for a line $\perp$ a plane needs:

one line

two intersecting lines of the plane

two parallel lines

three points

The three perpendiculars theorem relates:

two planes

an oblique and its projection

two obliques

two normals

A dihedral angle's arms must be $\perp$ to:

each other

the edge

the base

the normal

The shortest segment from a point to a plane is:

any oblique

the perpendicular

the projection

the diagonal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Prove $DD_1 \perp AC$

ANSWER $DD_1 \perp ABCD$ and $AC \subset ABCD$

2. Perpendicular 9, projection 12: the oblique

ANSWER 15

3. Perpendicular 9, projection 40: the oblique

ANSWER 41

4. Oblique 20 at 30° : the perpendicular

ANSWER 10

5. Same: the projection

ANSWER $10\sqrt{3}$

6. Pyramid base 10, height 12: slant height

ANSWER 13

7. Cube edge 6: distance from A to BDD_1B_1

ANSWER $3\sqrt{2}$

Medium · the standard question

7 PROBLEMS

1. Prove $AC_1 \perp BD$ by the three perpendiculars

ANSWER Projection AC , and $BD \perp AC$

2. Pyramid base 10, height 12: face-to-base angle

ANSWER $\approx 67.4^\circ$

3. Pyramid base 10, height 12: lateral edge

ANSWER $\sqrt{194} \approx 13.9$

4. Cube edge 6: distance from A to A_1BD

ANSWER $2\sqrt{3}$

5. Oblique 25, projection 24: the perpendicular

ANSWER 7

6. Prove $BCC_1B_1 \perp ABCD$

ANSWER $BB_1 \perp ABCD$ lies in it

7. Cube: angle of AC_1 with the base

ANSWER $\approx 35.26^\circ$

Hard · stretch and reason

7 PROBLEMS

1. Cube edge a : distance from A to the plane A_1BD

ANSWER $\frac{a\sqrt{3}}{3}$

2. Pyramid base 12, lateral edge 10: height and both dihedral angles

ANSWER $\sqrt{82}$, $\approx 56.4^\circ$ and $\approx 46.9^\circ$

3. Regular tetrahedron edge a : the dihedral angle between faces

ANSWER $\approx 70.5^\circ$

4. Cube edge 8: distance from B to the line A_1D_1

ANSWER $8\sqrt{2}$

5. A point 15 above a plane, two obliques at 30° and 45° : their lengths

ANSWER 30 and $15\sqrt{2}$

6. Prove the linear angle is independent of the point chosen on the edge

ANSWER Respectively parallel arms

7. Cube: prove $AC_1 \perp A_1BD$

ANSWER Two of its lines are $\perp$ to AC_1

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. Name the test on every proof.

1. Rewrite in full every question that lost a mark, naming the test used.
2. Five problems from the section your knowledge column was heaviest in.
3. A regular square pyramid has base edge 16 and height 15. Find the slant height, the lateral edge and both dihedral angles.

4. Cube of edge 10 : find the distance from A to the plane A_1BD .

Orthogonal projection and its technical applications

The drawing convention every engineer in the world uses — three views, and the object they determine.

LESSON 45–47

3 × 40 MIN

GEOMETRY 10, §18

IGCSE E14.1

LESSON CLOCK

12'

22'

26'

22'

26'

12'

Orthogonal versus general	12 min
The three views	22 min
Reconstructing a solid	26 min
The area formula	22 min
Practice	26 min
Homework	12 min

LEARNING OBJECTIVES

- Distinguish orthogonal from general parallel projection.
- Draw the plan, front and side elevations of a solid.
- Reconstruct a solid from its three views.
- Use the area formula for the orthogonal projection of a plane figure.

TEXTBOOK

National	pp. 211–228
Cambridge	Core & Extended, pp. 210–216
Workbook	Exercise 18.1

01

Explanation

Orthogonal versus general

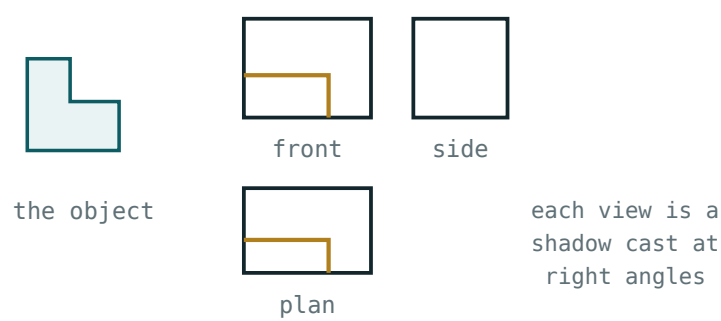
DEFINITION

Orthogonal projection is parallel projection in which the direction of projection is **perpendicular** to the plane of projection. Every point drops straight down onto the plane.

It is the special case that engineers use, because it preserves one thing a general parallel projection does not: **lengths parallel to the plane of projection are unchanged**. A drawing in orthogonal projection can therefore be measured — in that direction.

The projection of a point is the foot of the perpendicular; the projection of a segment is the segment between the feet; the projection of an oblique is what Lesson 35–37 called its projection.

The three views



three orthogonal projections determine the solid

*The plan from above, the front elevation from the front,
the side elevation from the side.*

VIEW	LOOKING	SHOWS
plan	from above	length and width
front elevation	from the front	length and height
side elevation	from the side	width and height

THE CONVENTIONS

Views are aligned: the plan sits directly below the front elevation, and the side elevation directly beside it, so that matching dimensions line up. Hidden edges are drawn dashed. Every view is drawn to the same scale.

Reconstructing a solid

Three views normally determine the solid — but not always. Two different solids can share all three views if the ambiguity is hidden inside.

PLAN	FRONT	SIDE	SOLID
circle	rectangle	rectangle	cylinder, upright
circle	triangle	triangle	cone
circle	circle	circle	sphere
square	square	square	cube
square	triangle	triangle	square-based pyramid

Reading a set of views is the everyday skill of a workshop; drawing them is the everyday skill of a design office. Both are the same theorem applied twice.

The area of a projection

THEOREM

If a plane figure of area S lies in a plane making a dihedral angle φ with the plane of projection, its orthogonal projection has area

$$S' = S \cdot \cos \varphi$$

Because lengths along the edge are unchanged and lengths perpendicular to it are multiplied by $\cos \varphi$, the area is scaled once.

Example. A square of side 10 lies in a plane at 60° to the horizontal. Its shadow on the horizontal has area $100 \cos 60^\circ = 50$.

THE ANGLE IS THE DIHEDRAL ANGLE

Not the angle of a line, and not the angle to the normal. If the figure's plane is at φ to the projection plane, the factor is $\cos \varphi$ — and $\varphi = 90^\circ$ collapses the figure to a segment, as it should.

02 Worked examples

EXAMPLE 1 A solid has a circular plan and rectangular front and side elevations. Name it.

- ① Circular from above.

- ② Rectangular from the front and side.

- ③ Constant circular cross-section.

ANSWER

An upright cylinder

EXAMPLE 2 A triangle of area 48 cm^2 lies in a plane at 30° to the horizontal. Find the area of its plan.

- ① $S' = S \cos \varphi$

- ② $= 48 \cos 30^\circ$

- ③ $= 48 \times 0.866 \approx 41.6$

ANSWER

$$24\sqrt{3} \approx 41.6 \text{ cm}^2$$

EXAMPLE 3 Cube of edge 6. Find the area of the orthogonal projection of the section ABC_1D_1 onto the base.

① The section is a rectangle $6 \times 6\sqrt{2}$, area $36\sqrt{2}$.

② It makes 45° with the base.

From Lesson 40–42.

③ $36\sqrt{2} \times \cos 45^\circ = 36$

The base square itself.

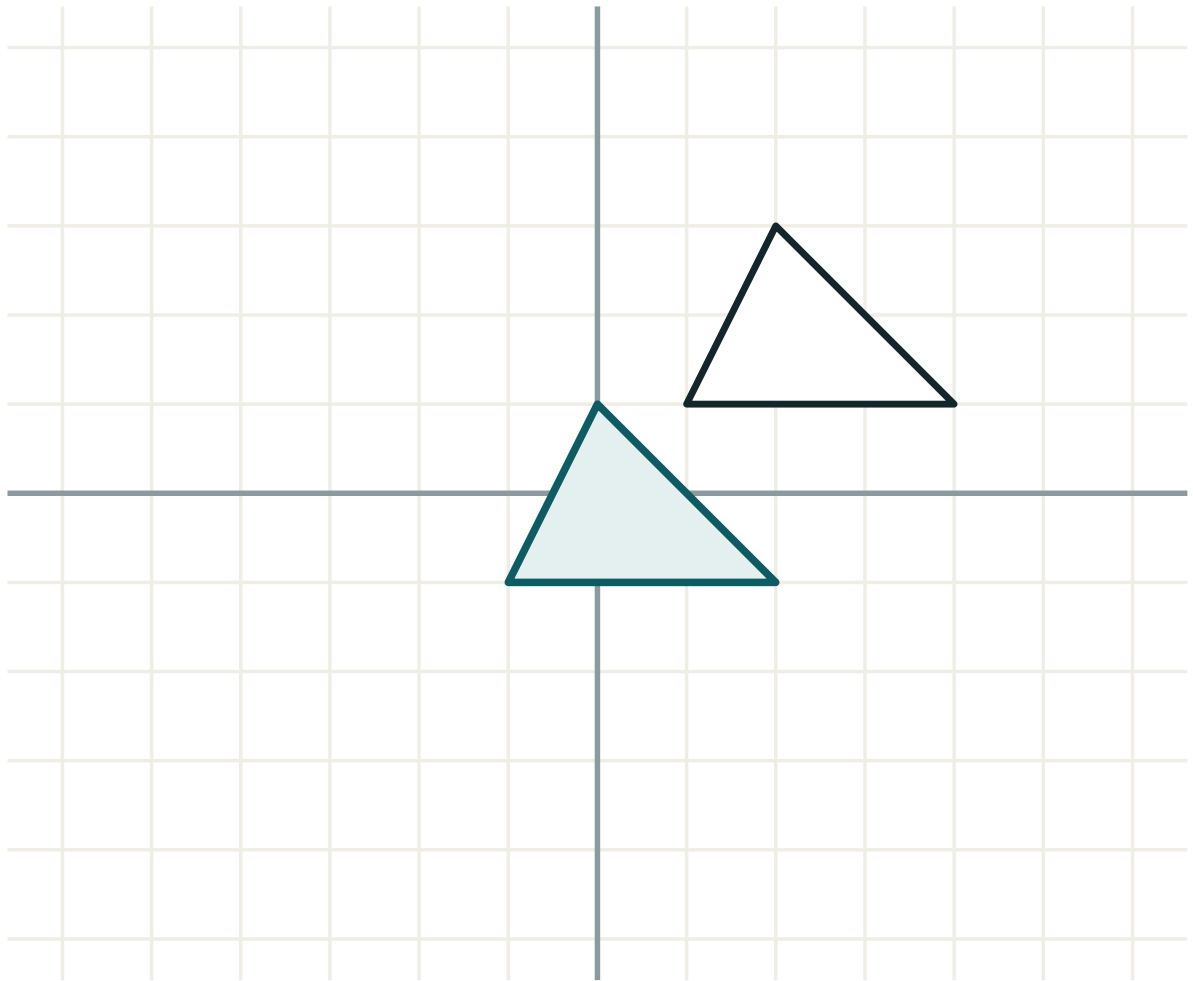
ANSWER

36 — the whole base

03 Interactive model

Shine a torch straight down on a tilted card and measure the shadow.

A projection



transformation **translation**

lengths **unchanged**

angles **unchanged**

orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Orthogonal projection	Ortogonal proyeksiya	Ортогональная проекция
Plan (top view)	Yuqoridan ko'rinish	Вид сверху
Front elevation	Old ko'rinish	Вид спереди
Side elevation	Yon ko'rinish	Вид сбоку
Technical drawing	Chizmachilik	Черчение
Hidden line	Ko'rinmas chiziq	Невидимая линия
Projection of an area	Yuza proyeksiyasi	Проекция площади
Scale	Masshtab	Масштаб
Isometric drawing	Izometrik chizma	Изометрия

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Orthogonal projection has the direction:

at any angle

perpendicular to the plane

along the plane

at 45°

The plan of a solid is the view from:

the front

the side

above

below

Circle, rectangle, rectangle is:

a cone

a cylinder

a sphere

a cube

A figure of area S at φ projects to:

$$S \sin \varphi$$

$$S \cos \varphi$$

$$S \tan \varphi$$

$$S$$

A square of side 10 at 60° projects to area:

$$50$$

$$86.6$$

$$100$$

$$25$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The plan is the view from

ANSWER above

2. The front elevation shows

ANSWER length and height

3. Hidden edges are drawn

ANSWER dashed

4. Circle, circle, circle is

ANSWER a sphere

5. Square, square, square is

ANSWER a cube

6. Circle, triangle, triangle is

ANSWER a cone

7. S' for $S = 20, \varphi = 60^\circ$

ANSWER 10

Medium · the standard question

7 PROBLEMS

1. Circle, rectangle, rectangle is

ANSWER an upright cylinder

2. Square, triangle, triangle is

ANSWER a square pyramid

3. Triangle of area 48 at 30° : its plan

ANSWER $24\sqrt{3}$

4. Square of side 10 at 45° : its plan area

ANSWER $50\sqrt{2} \approx 70.7$

5. A figure projects to half its area: the angle

ANSWER 60°

6. A figure projects to nothing: the angle

ANSWER 90°

7. Draw the three views of a cuboid $4 \times 3 \times 2$

ANSWER three rectangles: 4×3 , 4×2 , 3×2

Hard · stretch and reason

7 PROBLEMS

1. Cube edge 6: projection of the section ABC_1D_1 on the base

ANSWER 36

2. Cube edge a : projection of the hexagonal midpoint section on the base

ANSWER $\frac{3}{4}a^2$

3. A circle of radius 5 at 60° : its plan

ANSWER an ellipse of area 12.5π

4. A regular hexagon of side 4 at 45° : its plan area

ANSWER $12\sqrt{3} \times \frac{\sqrt{2}}{2} \approx 29.4$

5. Two different solids with the same three views: give an example

ANSWER A cube, and a cube with an internal cavity

6. Prove $S' = S \cos \varphi$ for a triangle with a side on the edge

ANSWER The height is scaled by $\cos \varphi$

7. A roof face of 60 m^2 at 35° : its plan area

ANSWER $\approx 49.1 \text{ m}^2$

07 Homework

Homework — 6 tasks

Draw the three views aligned, to one scale, with hidden edges dashed.

1. Draw the plan, front and side elevations of a cuboid $5 \times 3 \times 2$.
2. Draw the three views of a square-based pyramid and of an upright cylinder.
3. Name the solid with plan a circle, front a triangle and side a triangle.
4. A rectangle of area 72 cm^2 lies at 40° to the horizontal. Find the area of its plan.
5. Cube of edge 8: find the area of the projection of the section ABC_1D_1 on the base.

6. Explain in three sentences why an orthogonal projection can be measured but a general parallel projection cannot.

Bearings and angles of elevation in three dimensions

Cambridge insert: navigation and surveying, where the plan view and the vertical triangle are solved separately.

LESSON 48–49

2 × 40 MIN

GEOMETRY 10, §18 (EXTENSION)

IGCSE E14.2 · E6.6

LESSON CLOCK

9'

16'

18'

16'

15'

6'

Bearings	9 min
Elevation and depression	16 min
Working in the plan first	18 min
Combined problems	16 min
Practice	15 min
Homework	6 min

LEARNING OBJECTIVES

- Measure and use a three-figure bearing.
- Find the angle of elevation and depression in a 3-D setting.
- Solve a navigation problem by working in the plan first.
- Combine the plan triangle with the vertical triangle.

TEXTBOOK

National	pp. 229–234
Cambridge	Core & Extended, pp. 217–222, 305–312
Workbook	IGCSE Exercise 14.2

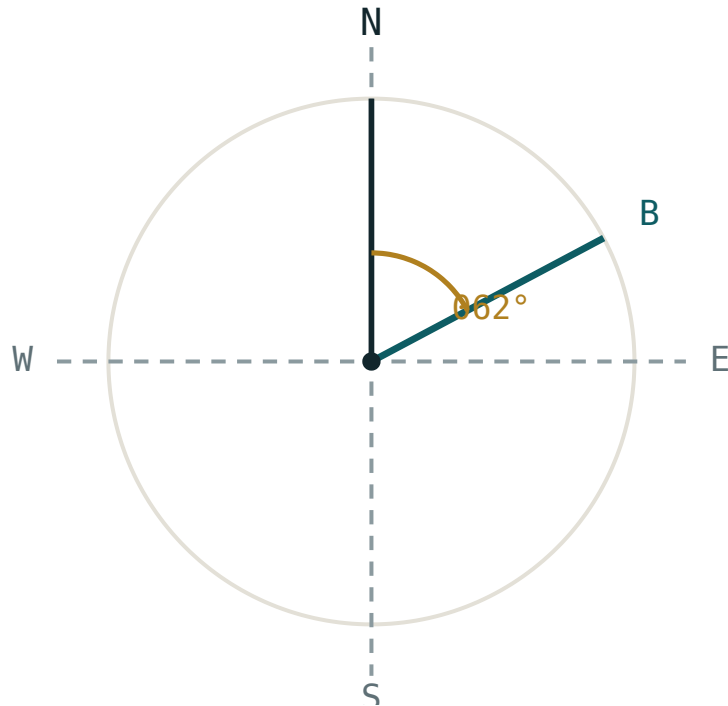
01

Explanation

Bearings

THE CONVENTION

A **bearing** is measured **from north, clockwise**, and written with **three figures**: 045° , 120° , 300° . Due east is 090° , south 180° , west 270° .



North at the top, clockwise, always three figures.

The **back bearing** — the bearing of A from B when the bearing of B from A is θ — is $\theta + 180^\circ$, reduced by 360° if it exceeds it.

THREE FIGURES, ALWAYS

45° is written 045° . It is not decoration: on a chart a missing leading zero has caused real navigational error.

Elevation and depression

The **angle of elevation** of a point is the angle its line of sight makes with the horizontal, measured upwards; the **angle of depression** is the same measured downwards.

angle of depression from A to B = angle of elevation from B to A

They are alternate angles between two horizontals — which is why a question may give either and expect the other.

Working in the plan first

THE METHOD FOR EVERY 3-D NAVIGATION PROBLEM

- 1** Draw the **plan** — a flat map — and solve it with the sine or cosine rule to find the horizontal distance.
- 2** Draw the **vertical triangle** containing that horizontal distance and the height. **3** Solve it with ordinary right-angle trigonometry.

Example. From A , a tower is on a bearing of 070° at 300 m; from B , 400 m due east of A , its bearing is 340° . The angle of elevation of the top from A is 9° . Find its height.

The plan gives the horizontal distance $AT = 300$ m directly, so the vertical triangle gives $h = 300 \tan 9^\circ \approx 47.5$ m. The second bearing is a check.

Combined problems

A full surveying problem chains three steps:

1. bearings $\rightarrow$ a plan triangle $\rightarrow$ the cosine or sine rule $\rightarrow$ the horizontal distance;
2. horizontal distance + angle of elevation $\rightarrow$ the height;
3. height + a second horizontal distance $\rightarrow$ a second angle.

Example. A mast stands at C . From A its elevation is 12° ; from B , 200 m from A on a bearing making 50° at C with AC , its elevation is 18° . With h the height, $AC = h \cot 12^\circ$ and $BC = h \cot 18^\circ$; the cosine rule in the plan then gives one equation in h alone.

NEVER MIX THE PLAN AND THE VERTICAL TRIANGLE

Bearings live in the plan; elevations live in the vertical. Solve one, carry the single number across, and solve the other. Trying to do both at once is where every mark is lost.

02 Worked examples

EXAMPLE 1 The bearing of B from A is 062° . Find the bearing of A from B .

① $62 + 180 = 242$

② Less than 360° , so no reduction.

ANSWER

242°

EXAMPLE 2 A tower is 300 m away horizontally; its top has elevation 9° . Find its height.

$$\textcircled{1} \quad \tan 9^\circ = \frac{h}{300}$$

$$\textcircled{2} \quad h = 300 \times 0.1584$$

$$\textcircled{3} \quad \approx 47.5 \text{ m.}$$

ANSWER

$\approx 47.5 \text{ m}$

EXAMPLE 3 From a cliff 80 m high, a boat has an angle of depression of 23° . Find its distance from the foot.

$\textcircled{1}$ The angle of elevation from the boat is also 23° .

$$\textcircled{2} \quad \tan 23^\circ = \frac{80}{d}$$

$$\textcircled{3} \quad d = \frac{80}{0.4245} \approx 188.4 \text{ m.}$$

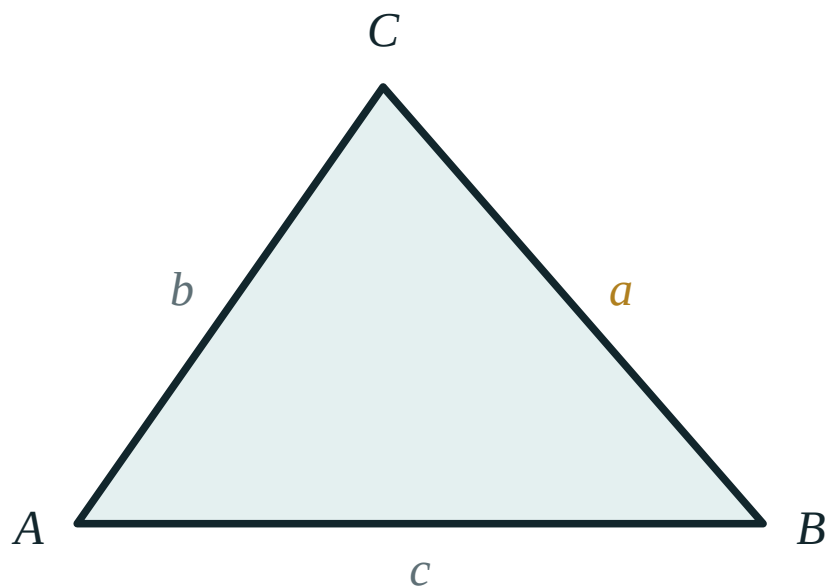
ANSWER

$\approx 188 \text{ m}$

03 Interactive model

Draw the plan on one half of the board and the vertical triangle on the other.

The plan triangle



b 7

c 9

A 55°

a 7.6

B 49° C 76°

area 25.8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Bearing	Azimut	Азимут
Three-figure bearing	Uch xonali azimuth	Трёхзначный азимут
Angle of elevation	Ko'tarilish burchagi	Угол возвышения
Angle of depression	Pasayish burchagi	Угол понижения
Plan view	Yuqoridan ko'rinish	Вид сверху
Due north	Shimolga	На север
Back bearing	Teskari azimuth	Обратный азимут
Line of sight	Ko'rish chizig'i	Линия визирования
Surveying	Geodeziya	Геодезия

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A bearing is measured from:

east, anticlockwise

north, clockwise

north, anticlockwise

south, clockwise

Due west is:

090°

180°

270°

360°

The back bearing of 062° is:

118°

242°

298°

062°

The angle of depression from A to B equals:

the elevation from B to A

its complement

nothing

twice it

In a 3-D navigation problem, solve first:

the vertical triangle

the plan

both at once

neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Bearing of due east

ANSWER 090°

2. Bearing of due south

ANSWER 180°

3. Bearing of due west

ANSWER 270°

4. Back bearing of 045°

ANSWER 225°

5. Back bearing of 300°

ANSWER 120°

6. Height with distance 100, elevation 30°

ANSWER ≈ 57.7

7. Distance with height 50, depression 45°

ANSWER 50

Medium · the standard question

7 PROBLEMS

1. Back bearing of 062°

ANSWER 242°

2. Tower 300 m away, elevation 9° : its height

ANSWER ≈ 47.5 m

3. Cliff 80 m, depression 23° : the distance

ANSWER ≈ 188 m

4. A mast of 40 m at 120 m: the angle of elevation

ANSWER $\approx 18.4^\circ$

5. A plane at 3000 m, depression to the airport 12° : the ground distance

ANSWER $\approx 14\ 100$ m

6. Bearing 130° for 5 km: how far east and how far south?

ANSWER 3.83 km east, 3.21 km south

7. From A, B is at 070° 4 km; C at 160° 3 km. Find BC

ANSWER 5 km

Hard · stretch and reason

7 PROBLEMS

1. A tower: elevations 12° and 18° from two points 200 m apart in line with it. Find its height

ANSWER ≈ 79.9 m

2. Same, with the two points and the tower forming a plan angle of 50°

ANSWER Cosine rule in $h \cot 12^\circ$ and $h \cot 18^\circ$

3. A ship sails 070° for 8 km then 160° for 6 km. Find its distance and bearing from the start

ANSWER 10 km on $\approx 107^\circ$

4. From a hill 120 m high, two boats have depressions 15° and 25° on the same bearing. Distance apart

ANSWER ≈ 190 m

5. A mountain has elevation 8° from 20 km and 11° from a nearer point. Find the height and the second distance

ANSWER ≈ 2.81 km; ≈ 14.5 km

6. A pylon is at 040° , 500 m from A and at 310° from B due east of A. Find AB

ANSWER ≈ 780 m

7. Explain why the plan is solved first

ANSWER It supplies the single horizontal distance the vertical triangle needs

07 Homework

Homework — 5 tasks

Draw the plan and the vertical triangle as two separate diagrams.

1. Give the back bearings of 035° , 145° and 290° .
2. A tower is 450 m away and its top has elevation 11° . Find its height.
3. From a cliff 95 m high a boat has depression 19° . Find its distance from the foot.
4. A ship sails 060° for 12 km, then 150° for 9 km. Find its distance and bearing from the start.

5. A mast: from two points 150 m apart in line with it the elevations are 14° and 22° . Find its height.

Revision of the chapter

One hour to draw the whole of perpendicularity into a single map before the projects.

LESSON 50

1 × 40 MIN

GEOMETRY 10, TAKRORLASH

IGCSE REVIEW

LESSON CLOCK

11'

12'

11'

6'

The tools, listed	11 min
Choosing a tool	12 min
The concept map	11 min
Setting the project	6 min

LEARNING OBJECTIVES

- Recall every test and theorem of the chapter without notes.
- Choose the right tool for a given statement.
- Build a concept map of perpendicularity in space.
- Identify the two weakest areas for the project week.

TEXTBOOK

National	pp. 235–238
Cambridge	Core & Extended, pp. 345–346
Workbook	Revision sheet R3

01

Explanation

The tools, listed

TOOL	STATEMENT	LESSON
test for $\ell \perp \alpha$	$\perp$ to two intersecting lines of α	33–34
Lemma 1	$a \parallel b, a \perp \alpha \implies b \perp \alpha$	33–34
Lemma 2	$a \perp \alpha, b \perp \alpha \implies a \parallel b$	33–34
shortest segment	the perpendicular beats every oblique	35–37
comparing obliques	longer projection $\iff$ longer oblique	35–37
three perpendiculars	$c \perp AB \iff c \perp OB$	38–39
test for $\beta \perp \alpha$	β contains a line $\perp$ to α	40–42
area of a projection	$S' = S \cos \varphi$	45–47

Choosing a tool

YOU ARE ASKED TO...	REACH FOR...
prove a line $\perp$ a plane	the test — two intersecting lines
prove two lines $\perp$ in space	three perpendiculars, or $\perp$ to a plane
prove two planes $\perp$	the test — one line $\perp$ to the other plane
find a distance to a plane	the perpendicular, then Pythagoras
find a distance to a line in space	three perpendiculars, then Pythagoras
find a dihedral angle	construct the linear angle
find a shadow's area	$S' = S \cos \varphi$

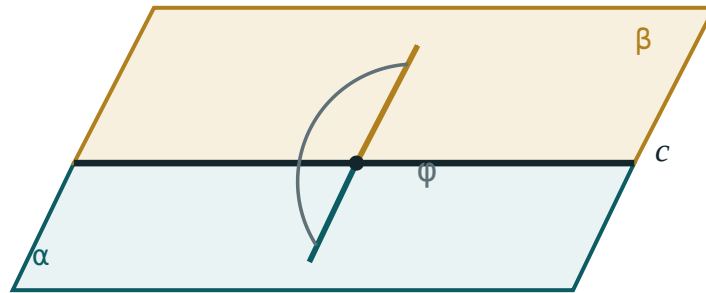
THE ONE HABIT

Almost every proof goes: **establish perpendicularity to a plane**, then read off perpendicularity to any line in it. If you are stuck, ask which plane you should be proving something about.

The concept map, and the project

Six boxes, links as sentences:

- **test** $\rightarrow \ell \perp \alpha$ — “two intersecting lines are enough”
- $\ell \perp \alpha \rightarrow$ **every line of α** — “that is the definition”
- **perpendicular** $\rightarrow$ **distance** — “the shortest segment”
- **oblique + projection** $\rightarrow$ **three perpendiculars** — “one right angle gives the other”
- **line $\perp$ plane** $\rightarrow$ **plane $\perp$ plane** — “contain the normal”
- **orthogonal projection** $\rightarrow S' = S \cos \varphi$ — “one direction is scaled”



both arms are perpendicular to the edge c
the angle φ between them is the dihedral angle

The picture behind two of the six boxes.

SETTING THE PROJECT

Each pair chooses one of: a scale model of a building with its three orthogonal views; a surveying exercise measuring a real height by two elevations; or a physical demonstration of the three perpendiculars theorem. Two lessons to prepare, presented next lesson.

02 Worked examples

EXAMPLE 1 Which tool proves $AC_1 \perp BD$ in a cube?

- ① Two lines in space, one oblique to the base.
- ② Its projection is AC .
- ③ $BD \perp AC$ in the base.

ANSWER

The theorem of the three perpendiculars

EXAMPLE 2 Which tool finds the distance from A to the plane BDD_1B_1 ?

- ① A distance to a plane is a perpendicular.
- ② $AC \perp BDD_1B_1$ by the test.
- ③ Half of AC .

ANSWER

The test for a line $\perp$ a plane, then Pythagoras

EXAMPLE 3 Which tool finds the shadow of a tilted square?

- ① An area under orthogonal projection.
- ② $S' = S \cos \varphi$.

ANSWER

The projection-area formula

03 Interactive model

Ask for the tool, not the answer, on ten statements in ten minutes.

The chapter in ten questions

The test for $\ell \perp \alpha$ needs:

one line

two intersecting lines

two parallel lines

three

Two lines $\perp$ to one plane are:

perpendicular

parallel

skew

equal

The shortest segment to a plane is:

an oblique

the perpendicular

the projection

the diagonal

Longer projection means:

shorter oblique

longer oblique

the same

nothing

Three perpendiculars relates:

two planes

an oblique and its projection

two obliques

two normals

To prove $\beta \perp \alpha$:

find a parallel line

find in β a line $\perp \alpha$

measure

use the definition

A dihedral angle's arms must be $\perp$ to:

each other

the edge

the base

the normal

Area under orthogonal projection:

$$S \sin \varphi$$

$$S \cos \varphi$$

$$S \tan \varphi$$

$$S$$

Perpendicular 9, projection 12: the oblique:

$$15$$

$$21$$

$$3$$

$$108$$

Pyramid base 10, height 12: slant height:

$$12$$

$$13$$

$$15$$

$$5$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Revision	Takrorlash	Повторение
Concept map	Tushunchalar xaritasi	Карта понятий
Test (criterion)	Alomat	Признак
Theorem	Teorema	Теорема
Dihedral angle	Ikki yoqli burchak	Двугранный угол
Projection	Proyeksiya	Проекция
Distance	Masofa	Расстояние
Self-assessment	O'z-o'zini baholash	Самооценка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The commonest proof shape in this chapter is:

measure and compare

prove $\perp$ to a plane, then read off

use coordinates

guess

A distance in space is always:

any segment

a perpendicular

the longest segment

a projection

The linear angle of a dihedral angle is:

unique

independent of the point on the edge

both

neither

The project options are:

one

two

three

four

06

Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Tool for “prove $AA_1 \perp ABCD$ ”

ANSWER the test for line $\perp$ plane

2. Tool for “prove $AC_1 \perp BD$ ”

ANSWER three perpendiculars

3. Tool for “prove $ABB_1A_1 \perp ABCD$ ”

ANSWER the test for perpendicular planes

4. Tool for a shadow's area

ANSWER $S' = S \cos \varphi$

5. Perpendicular 3, projection 4: the oblique

ANSWER 5

6. Two lines $\perp$ to one plane are

ANSWER parallel

7. The arms of a linear angle are $\perp$ to

ANSWER the edge

Medium · the standard question

7 PROBLEMS

1. Perpendicular 9, projection 40: the oblique

ANSWER 41

2. Pyramid base 10, height 12: the slant height

ANSWER 13

3. Same: the face-to-base dihedral angle

ANSWER $\approx 67,4^\circ$

4. Cube edge 6: distance from A to BDD_1B_1

ANSWER $3\sqrt{2}$

5. Cube: the dihedral angle along AB to ABC_1D_1

ANSWER 45°

6. A square of side 8 at 60° : its shadow area

ANSWER 32

7. Oblique 26 at 30° : the perpendicular

ANSWER 13

Hard · stretch and reason

7 PROBLEMS

1. Cube edge a : distance from A to A_1BD

ANSWER $\frac{a\sqrt{3}}{3}$

2. Regular tetrahedron edge a : the dihedral angle between faces

ANSWER $\approx 70.5^\circ$

3. Cube edge 6: shadow of the hexagonal midpoint section on the base

ANSWER 27

4. A mast: elevations 14° and 22° from points 150 m apart in line. Its height

ANSWER ≈ 88.4 m

5. Pyramid base 12, lateral edge 10: both dihedral angles

ANSWER $\approx 56.4^\circ$ and $\approx 46.9^\circ$

6. Prove $AC_1 \perp A_1BD$ in a cube

ANSWER Two of its lines are $\perp$ to AC_1

7. Explain why the linear angle does not depend on the point chosen

ANSWER Respectively parallel arms give equal angles

07 Homework

Homework — 3 tasks

The project sheet is due at the presentation lesson.

1. Write the eight tools of the chapter from memory, with their statements.
2. Finish the concept map with all six links written as sentences.
3. Prepare the project: a scale model with three views, a surveying measurement, or a demonstration of the three perpendiculars.

Presentation of project work

Two lessons in which the class does the teaching — and every mathematical claim is defended.

LESSON 51–52

2 × 40 MIN

GEOMETRY 10, LOYIHA ISHI

IGCSE E14.X

LESSON CLOCK

5'

45'

14'

11'

5'

The criteria	5 min
Presentations, five minutes each	45 min
Peer assessment	14 min
Whole-class review	11 min
The single lesson learned	5 min

LEARNING OBJECTIVES

- Present a piece of mathematical work clearly to an audience.
- Defend each measurement and each calculation under questioning.
- Assess another pair’s work against stated criteria.
- Identify one thing to do differently next time.

TEXTBOOK

National	pp. 239–242
Cambridge	Core & Extended, pp. 210–222
Workbook	Project sheet P3

01

Explanation

The criteria

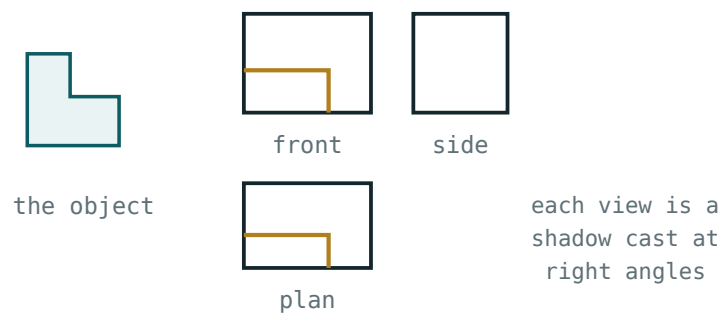
CRITERION	MARKS	WHAT EARNS THEM
the mathematics	6	correct theorems, correctly applied and named
the measurement or construction	4	done carefully, with the method described
the presentation	4	clear, audible, in five minutes, with a diagram
answering questions	4	defending a claim rather than repeating it
accuracy discussed	2	one honest sentence about the error

THE RULE FOR QUESTIONS

Every pair is asked at least two questions: one from the class and one from the teacher. At least one must be “*why is that angle a right angle?*” or “*how do you know that is the projection?*”

The three project types

TYPE	WHAT IS PRODUCED	THE MATHEMATICS TO DEFEND
scale model	a card building and its three orthogonal views	which lengths are preserved, and why
survey	a measured height from two angles of elevation	the plan triangle and the vertical triangle
demonstration	a physical model of the three perpendiculars	the statement, both directions



three orthogonal projections determine the solid

What a scale-model project has to produce, and get right.

Peer assessment, and the review

Each learner scores two other pairs on the five criteria and writes **one sentence** of specific praise and **one** specific suggestion. “Good work” scores nothing; “your plan view was aligned correctly under the elevation, which made it easy to read” scores full marks for the assessor.

THE WHOLE-CLASS REVIEW

Three questions, answered by the class together:

Which theorem was used most often across all the projects?

Which measurement was hardest to make accurately, and why?

Which presentation made its mathematics clearest, and what did it do?

Every learner then writes one line in the exercise book: “*Next time I will ...*”. Dated. It is read again at the end of Quarter IV.

02 Worked examples

EXAMPLE 1 A pair measured a tower as 47.5 m from 300 m and 9° . What should the accuracy sentence say?

- 1 The angle was measured to about $\pm 0.5^\circ$.
- 2 $300 \tan 8.5^\circ \approx 44.8$, $300 \tan 9.5^\circ \approx 50.2$.
- 3 So the height is between about 45 and 50 m.

ANSWER

"The height is 45–50 m; the angle dominates the error"

EXAMPLE 2 A pair claims their front elevation shows the true length of a sloping edge. Is that right?

- 1 Orthogonal projection preserves lengths **parallel** to the plane.
- 2 A sloping edge is not parallel to the front plane.
- 3 It appears foreshortened.

ANSWER

No — only lengths parallel to the plane of projection are preserved

EXAMPLE 3 A demonstration shows $c \perp OB$ and claims $c \perp AB$. What must be checked first?

- ① That AO really is perpendicular to the plane.
- ② That c lies in the plane and passes through B .
- ③ Only then does the theorem apply.

ANSWER

The two hypotheses of the theorem

03 Interactive model

Time each presentation strictly at five minutes; the discipline is part of the mark.

Assessing a project

Most marks are given for:

the model's appearance

the mathematics

the length of the talk

neatness

Orthogonal projection preserves lengths:

always

parallel to the plane

perpendicular to the plane

never

A survey needs:

one angle

two angles or one angle and a distance

three angles

no measurement

A useful peer comment is:

“good work”

specific about one thing

“I agree”

a mark only

An accuracy sentence should mention:

the answer

which measurement dominates the error

the weather

nothing

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Project work	Loyiha ishi	Проектная работа
Presentation	Taqdimot	Презентация
Criteria	Mezonlar	Критерии
Peer assessment	O'zaro baholash	Взаимооценка
Justification	Asoslash	Обоснование
Measurement	O'lchash	Измерение
Scale model	Masshtabli model	Масштабная модель
Survey	O'lchov ishi	Съёмка
Conclusion	Xulosa	Заключение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

How many questions must each pair answer?

none

at least two

exactly one

ten

The accuracy criterion is worth:

0

2

6

10

Peer assessment requires:

a mark only

one praise and one suggestion

nothing written

a full essay

The lesson ends with:

a test

one dated sentence in the exercise book

homework

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. How long is each presentation?

ANSWER five minutes

2. How many criteria are there?

ANSWER 5

3. Total marks

ANSWER 20

4. Marks for the mathematics

ANSWER 6

5. Marks for accuracy discussed

ANSWER 2

6. Are questions part of the mark?

ANSWER yes, 4

7. What is written at the end?

ANSWER one dated "next time I will ..." sentence

Medium · the standard question

7 PROBLEMS

1. A tower from 300 m at $9^\circ \pm 0.5^\circ$: the range of heights

ANSWER ≈ 45 to 50 m

2. Does the front elevation show a sloping edge's true length?

ANSWER no

3. Which lengths does orthogonal projection preserve?

ANSWER those parallel to the plane

4. What must be checked before the three perpendiculars theorem?

ANSWER $AO \perp \alpha$, and $c \subset \alpha$ through B

5. A scale model at 1:50: a 12 m wall is how long?

ANSWER 24 cm

6. Same model: a 60 m² wall has what model area?

ANSWER 240 cm²

7. Which theorem is used most across the projects?

ANSWER the three perpendiculars

Hard · stretch and reason

7 PROBLEMS

1. A survey gives $h = d \tan \theta$. Which of d or θ usually dominates the error?

ANSWER θ — the tangent is sensitive near steep angles

2. A model at 1:50 uses 0.2 litres of paint. The real building needs?

ANSWER 500 litres

3. Show that a 0.5° error at 9° changes the height by about 6%

ANSWER $\frac{\Delta h}{h} \approx \frac{\Delta \theta}{\sin \theta \cos \theta}$

4. Why is a demonstration of the converse harder to build than the forward direction?

ANSWER It must start from a right angle you cannot see

5. Design a check that a measured height is right

ANSWER Measure from a second point and compare

6. A projection of a 60 m^2 roof at 35° : the plan area

ANSWER $\approx 49 \text{ m}^2$

7. Give one specific peer comment on a set of three views

ANSWER e.g. “your plan is aligned under the elevation, so the widths match”

07 Homework

Homework — 3 tasks

The dated sentence is read again at the end of Quarter IV.

1. Write your peer assessment of two projects: the five scores, one specific praise and one specific suggestion each.
2. Write your own “next time I will ...” sentence, dated.
3. Answer the three whole-class review questions in your own words.